

String

✓ Array / group / sequence of characters
✗ not ordered ✓ ordered

"Hello World"

A to Z

a to z

0 to 9

. , ; : { [] } ...

"abc_08@" → String

"abc → Not a string (No double quotes at the end)

" → String

"123" → String

123 → Not a string

'123' → X

'5' → } character, not string

'x' → }

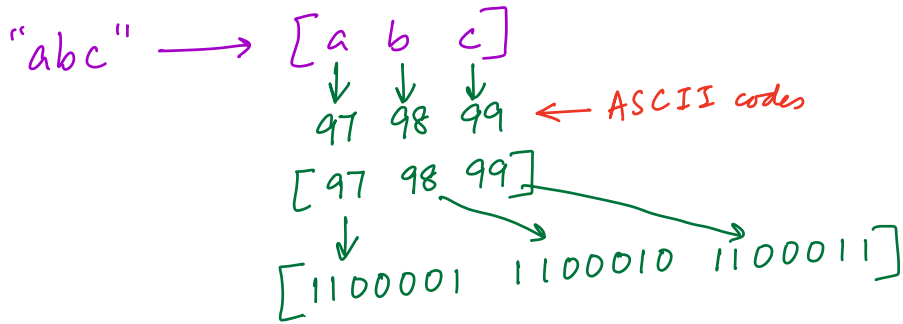
String vs Integer

"123" → String → supports string operations (like concatenation)

123 → integer → supports arithmetic / logical operations

ASCII code

In computer, everything is stored as bits (binary digits $\rightarrow 0, 1$)



"abc" + "d"

$\rightarrow$ [97 98 99 100]

'A' to 'Z'	'a' to 'z'	'0' to '9'
'A' $\rightarrow$ 65	'a' $\rightarrow$ 97	'0' $\rightarrow$ 48
'B' $\rightarrow$ 66	'b' $\rightarrow$ 98	'1' $\rightarrow$ 49
'C' $\rightarrow$ 67	'c' $\rightarrow$ 99	'2' $\rightarrow$ 50
$\vdots$	$\vdots$	$\vdots$
'Z' $\rightarrow$ 90	'z' $\rightarrow$ 122	'9' $\rightarrow$ 57

' ' (White space character) $\rightarrow$ 32 .

int x = '0';
sizeof(x); $\rightarrow$ 48 .

int y = 49; $\rightarrow$ '1'
sizeof((char) y); $\rightarrow$ 1

int x = 'A';
sizeof(x); $\rightarrow$ 65

'a' - 0 $\rightarrow$ 97 - 0
 $\uparrow \quad \uparrow$
character int
char $\quad$ = 97
 $\quad$ =

$$'a' - '0' = 97 - 48 \\ = 49$$

$$97 - '0' = 97 - 48 \\ = 49$$

$$'a' \% '0' = 97 \% 48 \\ = 1$$

$$'a' + '0' \rightarrow 97 + 48 \\ = 145$$

$$"a" + '0' \rightarrow "a0"$$

String char

$$" " + 'a' \rightarrow " a"$$

$$\text{char } a = 'a' - 'A';$$

$\downarrow \quad \quad \downarrow$
 $97 - 65 \rightarrow 32 \rightarrow 'b'$

String country = "India";

0	1	2	3	4
'I'	'n'	'd'	'i'	'a'

← array of characters

size $\rightarrow$ country.length();

$\uparrow$ *

System.out.println(country.length()); $\rightarrow 5$

ith character $\rightarrow$ country.charAt(i);

$\uparrow$ index

System.out.println(country.charAt(3)); $\rightarrow i$

Q) Given a string, print its characters in new lines

"India" →
I
n
d
i
a

```
fn (int i=0; i < str.length(); i++) {  
    Sopl(str.charAt(i));  
}
```

$\text{Sopl}('a' + '0' + "b");$
 $\xrightarrow{97 + 48} 145$

$145 + "b"$
↓
 $145b$

Q) Given a string, print the ASCII codes of its characters in new lines

"India"

73
110
100
105
97

```
fn (int i=0; i < str.length(); i++) {  
    Sopl((int) str.charAt(i));  
}  
OR  
Sopl(str.charAt(i) + 0);
```

[Break till 10:20 PM]

HashMap

Key-value store

states and their populations:-

```
HashMap<String, Integer> hm  
= new HashMap<>();
```

String (Key) <u>states</u>	int / long (Value) <u>Population</u>
Punjab	15
Haryana	12
Karnataka	13
UP	18

P → 15

H → 12

K → 13

AP → 13

UP → 18
 → 19 X

Key duplication
is not allowed.

```
HashMap<String, Integer> hm = new HashMap<>();
```

```
// Insert
```

```
hm.put("Punjab", 15);
```

```
hm.put("Haryana", 12);
```

```
hm.put("UP", 18);
```

```
System.out.println(hm);
```

```
P → 15  
H → 12  
UP → 18
```

```
// Remove by Key
```

```
hm.remove("Haryana");
```

```
System.out.println(hm);
```

```
P → 15  
UP → 18
```

```
System.out.println(hm.size());
```

```
// get
```

```
System.out.println(hm.get("UP"));
```

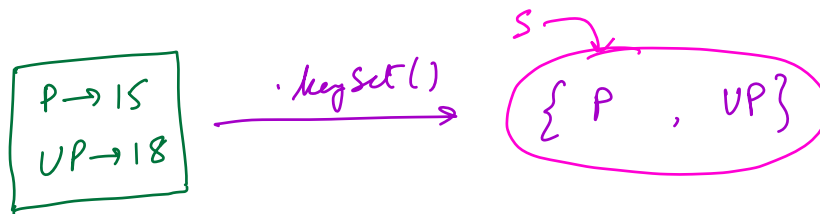
```
// check
```

```
System.out.println(hm.containsKey("Haryana"));
```

```
if (hm.containsKey("UP"))
```

```
System.out.println(hm.get("UP"));
```

```
for (String s : hm.keySet()) {  
    System.out.println(s + " → " + hm.get(s));  
}
```



$S \rightarrow P$

$S \rightarrow UP$

`ArrayList<Integer> an = ...
{3, 8, 6};`

`for (int x: an) {
 solve(x);
}`

$x=3$
 $x=8$
 $x=6$

Q.) Given an integer array, return the corresponding frequency map.

arr $\rightarrow \{1, 4, 3, -2, 1, 1, 4, 5, 3\}$

hm $\rightarrow \left[\begin{array}{l} 1 \rightarrow 3 \\ 3 \rightarrow 2 \\ 4 \rightarrow 2 \\ 5 \rightarrow 1 \\ -2 \rightarrow 1 \end{array} \right]$

$\{1, 4, 3, -2, 1, 1, 4, 5, 3\}$

HM

1	:	X 3
4	:	X 2
3	:	X 2
-2	:	1
5	:	1

```
HashMap<Integer, Integer> freq = new HashMap<>();
```

```
for(int i=0; i<arr.length; i++){
```

```
    if(!freq.containsKey(arr[i])){
```

```
        freq.put(arr[i], 1);
```

```
    }
```

```
    else{
```

```
        int v = freq.get(arr[i]);
```

```
        freq.put(arr[i], v+1);
```

```
    }
```

```
}
```

```
        freq.put(arr[i],  
            freq.get(arr[i])+1);
```



```
HashMap<Integer, Integer> freq = new HashMap<>();  
for (int i=0; i< ar.length; i++) {  
    hm.put(ar[i], 1+hm.getOrDefault(ar[i], 0));  
}
```

HashSet → collection of unique elements (keys)

